

— CRASH-COURSE WORKBOOK

Acids & Bases

the CARDIO ladder

One move ranks every acid on the DAT: **acidity = how stable is the conjugate base**. This workbook is yours to mark up — fill the blanks live, work the six, and keep the rules you write.

Name _____

Date _____

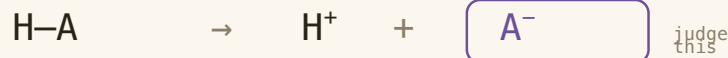
HOW TO USE THIS

Follow along in the live session and **write in it**. Do the six questions cold before checking the key. Every rule you write in your own words is one you keep. Bulk drilling lives in the Ace Labs engine — this is where you learn the move.

Forget the acid. Judge what it leaves behind.

Acidity = **how stable is the conjugate base?**

An acid gives up a proton (H^+). What's left is the **conjugate base**, holding the leftover negative charge. A strong acid is one whose leftover anion is *comfortable* holding that charge.



THE PICTURE — HOT POTATO

Treat the charge like a hot potato. The more hands it can be passed to — a bigger atom, resonance, a nearby group tugging — the comfier the anion, and the stronger the acid. **Crammed on one atom = unstable = weaker acid.**
Spread out = stable = stronger acid.

TRAP #1

Judge the **conjugate base**, never the neutral acid. Pull the proton, draw the anion, *then* ask how stable it is.

In your words: what does “stable conjugate base” mean?

Six letters, really five ideas: **CARDIO**

Walk two anions letter by letter. The **first real difference** is your answer — and your explanation. Fill the last column with your own example as we go. (D and I are one idea, two doorways.)

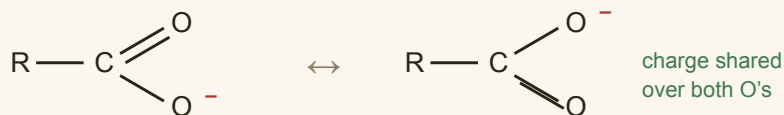
	REASON	THE RULE	YOUR EXAMPLE
C	Charge	Spread or neutral beats a full charge on one atom. A cationic acid (neutral conjugate base) is much stronger than its neutral relative.	
A	Atom	Which atom holds the charge? Across a row the more electronegative wins; down a column the bigger atom wins.	
R	Resonance	Spread the charge over several atoms. The biggest lever on the DAT.	
D	Dipole	A polar bond's field steadies the charge through space (the field half of D/I).	
I	Induction	Electron-withdrawing groups drain charge through the bonds. Closer and more groups = stronger.	
O	Orbital	More s-character holds the pair tighter. The last tiebreaker: $sp > sp^2 > sp^3$.	

THE HABIT

When a ranking question appears, draw each conjugate base and **say the letters out loud**. Stop at the first letter that separates them — that's the answer and the reason.

Resonance is the one to see once.

A carboxylate's charge lives half on each of two identical oxygens — far comfier than a full charge on one. That's why acetic acid (~5) crushes ethanol (~16) from the same O–H bond.



The pK_a anchors — know where these live



Left = strong (loves to give up H^+); right = barely acidic. Lower pK_a = stronger acid.

YOUR REP · PLACE THEM

Where would **ammonium (NH_4^+)**, **ethanol**, and **trichloroacetic acid** sit on the ladder above? Mark an arrow for each and write its pK_a .

Now you hold the pen.

Deprotonate each alcohol below. Draw the conjugate base (the anion) in each box. Then decide: **which anion can spread its charge?** That one came from the stronger acid.

Ethanol $\text{CH}_3\text{CH}_2\text{OH}$

DRAW THE ANION

Acetic acid CH_3COOH

DRAW THE ANION

Which anion spreads its charge? _____

So the stronger acid is _____ because _____

CHECK YOURSELF

Acetate spreads its charge over two equivalent oxygens by resonance; ethoxide has none. Acetic acid wins — by over a hundred billion times. Same O–H bond, different anion. That’s the whole topic in one pair.

Scratch space

The six. Commit before you check.

One for each CARDIO letter. Circle your answer and rate how sure you are — then check the key on the last page. A committed guess is what makes it stick.

Q1

C · CHARGE

A cationic acid tends to be much stronger than its neutral relative, because its conjugate base comes out neutral. Which is the strongest acid?

- ☐ (A) NH_4^+ (ammonium)
- ☐ (B) NH_3 (ammonia)
- ☐ (C) CH_4 (methane)
- ☐ (D) CH_3CH_3 (ethane)
- ☐ (E) CH_3NH_2 (methylamine)

☐ Sure ☐ Shaky ☐ No idea

why: _____

Q2

A · ATOM (ROW)

Each loses one proton. Across the second row of the periodic table, which is the strongest acid?

- ☐ (A) HF
- ☐ (B) H_2O
- ☐ (C) NH_3
- ☐ (D) CH_4
- ☐ (E) They are all equally strong

☐ Sure ☐ Shaky ☐ No idea

why: _____

Q3

A · ATOM (COLUMN)

Which hydrohalic acid is the strongest: HF, HCl, HBr, or HI?

- ☐ (A) HI
- ☐ (B) HF
- ☐ (C) HCl
- ☐ (D) HBr
- ☐ (E) All equally strong (all strong acids)

☐ Sure ☐ Shaky ☐ No idea

why: _____

Q4

R · RESONANCE

Acetic acid (pK_a 4.8) is over a hundred billion times more acidic than ethanol (16), though both lose an O–H proton. The main reason is that the acetate anion is:

- (A) delocalized — its charge is shared over two equivalent oxygens by resonance
- (B) sitting on a larger atom than ethoxide is
- (C) held right next to an sp-hybridized carbon
- (D) neutral overall, unlike ethoxide
- (E) stabilized by an internal hydrogen bond

☐ Sure ☐ Shaky ☐ No idea

why: _____

Q5

I · INDUCTION

Trichloroacetic acid (pK_a 0.7) is far stronger than acetic acid (4.8). Both carboxylates have identical resonance. The extra stabilization comes from:

- (A) three electron-withdrawing chlorines pulling charge away (induction)
- (B) extra resonance from the chlorine lone pairs
- (C) the greater molecular weight
- (D) chlorine raising the s-character of the carbon
- (E) hydrogen bonding between the Cl and the O–H

☐ Sure ☐ Shaky ☐ No idea

why: _____

Q6

O · ORBITAL

Which C–H bond is the most acidic?

- (A) the terminal C–H of ethyne (an alkyne)
- (B) a C–H of ethane (an alkane)
- (C) a C–H of ethene (an alkene)
- (D) all C–H bonds are equally acidic
- (E) ethane, because it has the most C–H bonds

☐ Sure ☐ Shaky ☐ No idea

why: _____

The six, decided.

- Q1 → A** **Charge.** Ammonium loses H^+ to give neutral NH_3 — reaching neutral is a big stabilization ($\text{pK}_a \sim 9$), so it beats every neutral N–H and C–H acid here.
-
- Q2 → A** **Atom, across a row.** The most electronegative atom holds the charge best: $\text{F} > \text{O} > \text{N} > \text{C}$. HF wins (3.2 vs 15.7 vs 38 vs ~ 50).
-
- Q3 → A** **Atom, down a column.** Size beats electronegativity going down; big iodide spreads the charge, so HI is strongest (~ -10). HF is actually the *weakest* halide — the classic trap.
-
- Q4 → A** **Resonance.** Half a charge on each of two equivalent oxygens beats a full charge on one. Ethoxide has no such escape.
-
- Q5 → A** **Induction.** Same resonance in both, so R ties. The three Cl atoms drain charge off the anion through the bonds and space (the D/I check) — not resonance or mass.
-
- Q6 → A** **Orbital.** The sp carbon is 50% s-character, holding the pair close (~ 25). More s-character = more acidic: $\text{sp} > \text{sp}^2 > \text{sp}^3$.
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YOUR RULES TO KEEP

Write, in your own words, the one-line rule behind each miss — then drill it in the Ace Labs engine.

Cut it out. Carry it.

When a ranking question appears, run this top-down and stop at the first real difference — that's the answer and the reason.

ACID-BASE CARDIO

Acidity = how stable is the conjugate base?

1. **C**harge — spread or neutral is better
2. **A**tom — row: EN wins; column: size wins
3. **R**esonance — delocalize the charge
4. **D/I** — nearby EWG pulls it away (closer = stronger)
5. **O**rbital — $sp > sp^2 > sp^3$

WHERE IT GOES FROM HERE

Now: drill CARDIO to reflex in the Ace Labs Acid-Base engine (timed sets + spaced deck). **In 2 days:** redo the six from memory before checking. **Re-test:** catch it on your next full length — land it there and it's yours.